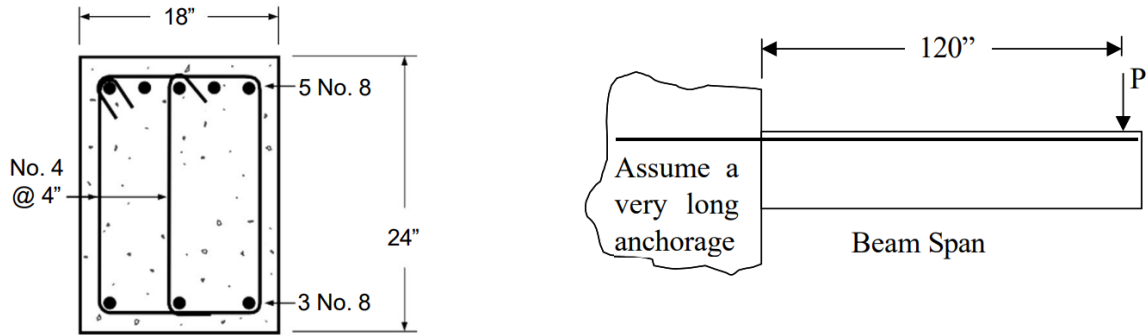


## 1 Preface

The beam cross section analyzed in Homework 4 is used in a weightless cantilever shown in **Fig. 1**. From that problem set, assume that the depth of the flexural compression zone after cracking but before yielding is 6.7 inches. Recall that the reinforcement yield stress is 69 ksi. Values of moment and curvature at relevant points are tabulated below.



**Figure 1:** Upside-down beam cross section diagram to be used.

| Point          | Description                             | Value / Calculation             | Units      |
|----------------|-----------------------------------------|---------------------------------|------------|
| $M_0$          | Initial condition (no applied load).    | $M = 0 \quad \phi = 0$          | k-in, 1/in |
| $M_{cr}$       | Onset of flexural cracking in concrete. | $M = 820 \quad \phi = 0.000011$ | k-in, 1/in |
| $M_y$          | Yielding of longitudinal reinforcement. | $M = 5240 \quad \phi = 0.00016$ | k-in, 1/in |
| $M_{spalling}$ | Concrete spalling at compression face.  | $M = 5370 \quad \phi = 0.0012$  | k-in, 1/in |

**Table 1:** Summary of key load and curvature values for the cantilever beam.

### 1.1 Load–Deflection Calculations

The load–deflection behavior of relatively slender beams, columns, and walls can be evaluated using classical flexural theory that accounts for concrete cracking and section yielding [1, 2, 3, 4, 5]. The general procedure consists of determining the curvature distribution along the member and integrating it to obtain rotations and displacements. For *statically determinate* members, the moment diagram is directly defined by the applied loading, leading to a straightforward derivation of the curvature distribution. In contrast, *statically indeterminate* structures exhibit internal moments that depend on both loading and stiffness, which requires an iterative, step-by-step solution where stiffness is updated after each load increment. This is not covered in this assignment.

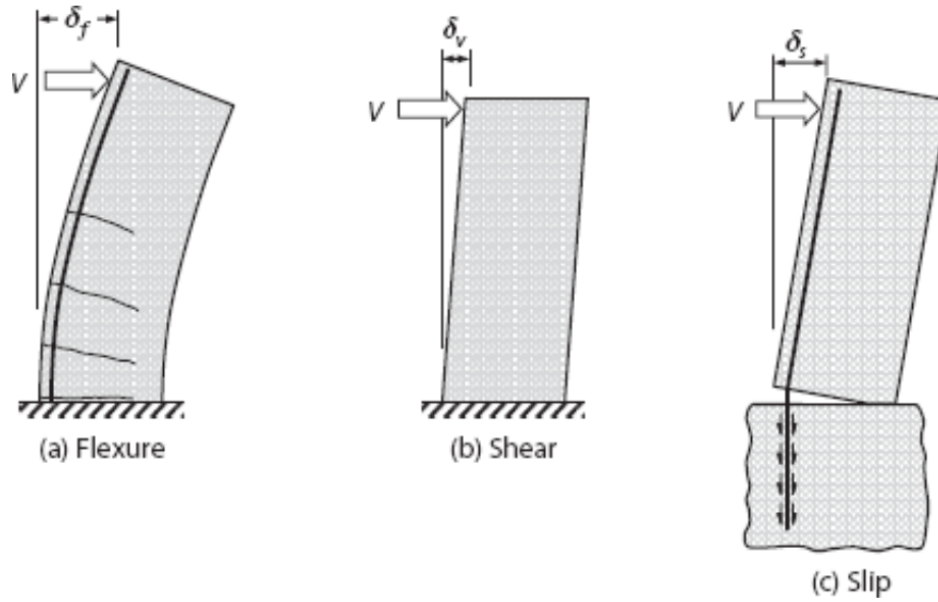
Flexural curvature alone may not fully describe the flexibility of a beam–column. In practice, additional deformation sources—such as shear distortion and reinforcement slip at anchorage zones—must also be considered. In inelastic regimes, the coupling between flexural and shear deformations may further influence the overall response.

In this homework, the objective is to compute the **load–deflection relationship** for a statically determinate beam, first analyzing its *linear response* and subsequently its *inelastic behavior*.

### 1.1.1 Linear Response

Linear response describes the structural behavior prior to yielding of either the tensile longitudinal reinforcement or the flexural compression zone [6]. Within this range, the total flexibility is decomposed into three primary components (refer to **Fig. 2**). The *flexural component* accounts for the displacement arising from flexural curvature, the *shear component* represents the deformation associated with conventional shear distortion, and the *slip component* corresponds to the rigid-body movement produced by reinforcement slip at the anchorage zones. The total displacement is therefore obtained as the sum of these three contributions, as shown in **Eq. (1)**.

$$\delta = \delta_f + \delta_v + \delta_s \quad (1)$$



**Figure 2:** Components of flexibility in beam-columns. (a) Flexure, (b) Shear, (c) Slip.

## 2 Problem a: Uncracked section

### Assignment 1

Calculate the displacement at the tip just before onset of cracking. Include flexure and shear (for the latter, because it is an uncracked beam, assume  $G_{\text{eff}} = 0.4E_c$ ). Because the beam is uncracked, assume zero slip from the anchorage.

### 2.1 Comment on flexural

For the uncracked section, a linearly static model for the Euler-Bernoulli beam theory is appropriate for calculating the total deflection. This is defined in **Eq. (3)**:

#### 2.1.1 From curvature to displacement

Based on the moment-curvature relationship, the flexural deformation of the cantilever beam can be obtained by integrating the curvature distribution along its length. For a cantilever of length  $l$  subjected to a tip load  $V$ , the curvature varies linearly from zero at the free end to its maximum value at the fixed support. The total flexural displacement at the tip is given by:

$$\delta_f = \int_0^l m_x(x) \phi_x dx \quad (2)$$

Where  $m_x(x)$  is the bending moment due to a unit load applied at the free end, and  $\phi_x(x)$  is the curvature caused by the actual loading.

This formulation is derived from the *principle of virtual work*, which states that the external virtual work performed by a unit load is equal to the internal virtual work associated with the corresponding stress (or curvature) distribution within the structure. In this context, the unit load generates a *virtual moment field*  $m_x(x)$ , while the real loading produces the *real curvature field*  $\phi_x(x)$ . Their product, integrated along the member length, gives the total tip displacement due to flexure.

#### 2.1.2 Linear Case

If the curvature distribution linear, the above relation simplifies to the classical expression:

$$\delta_{f,y} = \frac{\phi_y l^2}{3} \quad (3)$$

where  $\phi_y$  is the curvature at the fixed end corresponding to the yield moment.

### 2.2 Comment on shear

Shear deformations can be approximated by modeling the beam-column as a homogeneous, isotropic material with a constant shear modulus. For a cantilever, the corresponding shear displacement at the tip is given by [6]:

$$\delta_v = \frac{VL}{A_v G_{\text{eff}}} \quad (4)$$

Where  $A_v$  is the effective shear area and  $G_{\text{eff}}$  the effective shear modulus. Prior to shear cracking,  $G_{\text{eff}}$  may be approximated as  $0.4E_c$ , while after cracking it typically decreases. Following the recommendation by Elwood and Eberhard (2009), this study adopts  $G_{\text{eff}} = 0.2E_c$  for typical beams and columns.

### 2.3 Comment on slip

For a uncracked section, no slip is expected whatsoever. The bond relations between concrete and steel account for no different in their linear strain field.

### 2.4 Results

| Parameter               | Description                                       | Value / Calculation | Reference                          |
|-------------------------|---------------------------------------------------|---------------------|------------------------------------|
| $L$                     | Longitude of bar.                                 | 120 in              | <b>Fig. 1.</b>                     |
| $E_c$                   | Young elastic modulus.                            | 3605 ksi            | ACI-318[6].                        |
| $G$                     | Shear elastic modulus.                            | $0.4E_c = 1442$ ksi | Assumed for uncracked concrete     |
| $A_v$                   | Effective shear area of the column cross section. | 360 in <sup>2</sup> | EQ 6.42[6].                        |
| $f_{\text{max}}$        | Expected yield strength for steel.                | 69 ksi              | ACI 318 [7].                       |
| $\delta_f$              | Flexural displacement.                            | 0.05280 in          | From curvature integration. Linear |
| $\delta_v$              | Shear displacement.                               | 0.00158 in          | EQ 6.42[6]                         |
| $\delta_s$              | Slip displacement.                                | 0 in                | No slip in uncracked section.      |
| $\delta_{\text{total}}$ | Total displacement.                               | <b>0.05438</b> in   | Sum of components.                 |

**Table 2:** Flexural, shear, and slip displacement components for the uncracked section.

### 3 Problem b: Onset of Yielding: Cracked & Uncracked

#### Assignment 2

Calculate the displacement corresponding to onset of longitudinal reinforcement yielding, accounting for effects of curvature variation along the length and accounting for shear deformations (assume  $G_{\text{eff}} = 0.2E_c$ ), but not including slip from the anchorage. Make an accurate accounting of the variation of curvature along the length considering effects of uncracked sections along portions of the beam length for which moment is less than cracking moment.

#### 3.1 Comment on Flexural

To obtain the flexural displacement, the procedure described in **Eq. (2)** is followed. The integral is computed by the sum of areas in the curvature diagram showned in **Fig. 3**. The result is:

| Parameter  | Description                            | Value / Calculation | Reference                           |
|------------|----------------------------------------|---------------------|-------------------------------------|
| $A_1$      | Area 1 in <b>Fig. 3</b> .              | 0.000103282         | From curvature diagram integration. |
| $A_2$      | Area 2 in <b>Fig. 3</b> .              | 0.001113435         | From curvature diagram integration. |
| $A_3$      | Area 3 in <b>Fig. 3</b> .              | 0.007540992         | From curvature diagram integration. |
| $m_{x1}$   | Moment $m_x$ value at Area 1 centroid. | 12.52 kip – in      | Virtual moment distribution.        |
| $m_{x2}$   | Moment $m_x$ value at Area 2 centroid. | 69.39 kip – in      | Virtual moment distribution.        |
| $m_{x3}$   | Moment $m_x$ value at Area 3 centroid. | 86.26 kip – in      | Virtual moment distribution.        |
| $\delta_f$ | Total flexural displacement.           | <b>0.729</b> in     | $\sum(A_i m_{xi})$                  |

**Table 3:** Flexural displacement computation using the principle of virtual work.

#### 3.2 Comment on Shear

Similarly to Section 2.2, the shear modulus for the uncracked section should be replaced by  $0.20E_c$  and the moment to consider is  $M_y$ .

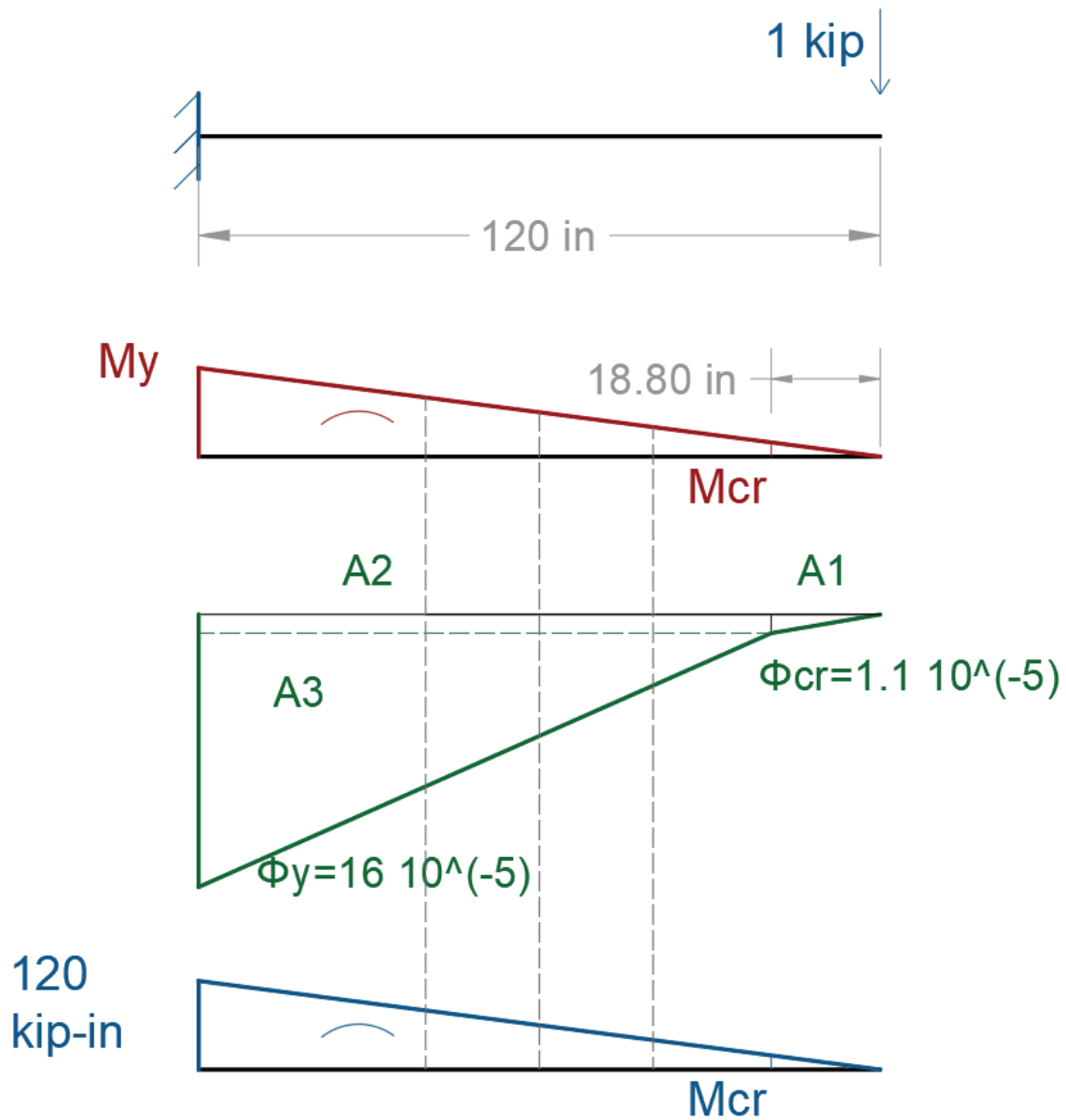
#### 3.3 Comment on Slip

The behavior is analogous to Section 2.3.

### 3.4 Results

| Parameter               | Description                                       | Value / Calculation | Reference                                  |
|-------------------------|---------------------------------------------------|---------------------|--------------------------------------------|
| $L$                     | Longitude of bar.                                 | 120 in              | Given                                      |
| $G$                     | Shear elastic modulus.                            | 721 ksi             | $0.2E_c$ , Cracked section.                |
| $A_v$                   | Effective shear area of the column cross section. | $360 \text{ in}^2$  | EQ 6.42[6].                                |
| $f_{\max}$              | Expected yield strength for steel.                | 69 ksi              | ACI 318 [7].                               |
| $\delta_f$              | Flexural displacement.                            | 0.729 in            | From curvature integration in Section 3.1. |
| $\delta_v$              | Shear displacement.                               | 0.0202 in           | Based on effective shear modulus.          |
| $\delta_s$              | Slip displacement.                                | 0.00000 in          | No slip considered due to assignment.      |
| $\delta_{\text{total}}$ | Total displacement.                               | <b>0.74922 in</b>   | $\delta_f + \delta_v + \delta_s$           |

**Table 4:** Flexural, shear, and slip displacement components for the cracked and uncracked section at the onset of yielding.



**Figure 3:** Bending moment and curvature diagram along beam span.

## 4 Problem c: Fully cracked beam.

### Assignment 3

Repeat the previous calculation but assume that the beam is fully cracked along the entire length, even at sections for which the moment is less than the cracking moment. (In other words, curvature at any point along the length of the beam is  $M/EI_{\text{eff}}$ , where  $M$  is the moment at that point along the beam.)

The approximation described in the problem statement is illustrated in **Fig. 4**, superimposed on **Fig. 3** for comparison. In this case, the entire section is assumed to be fully cracked, resulting in a constant beam stiffness along the entire span.

### 4.1 Comment on Shear and Slip

Shear and Slip contribution remains unchanged and follow the same procedure as Section 3.2 and Section 3.3.

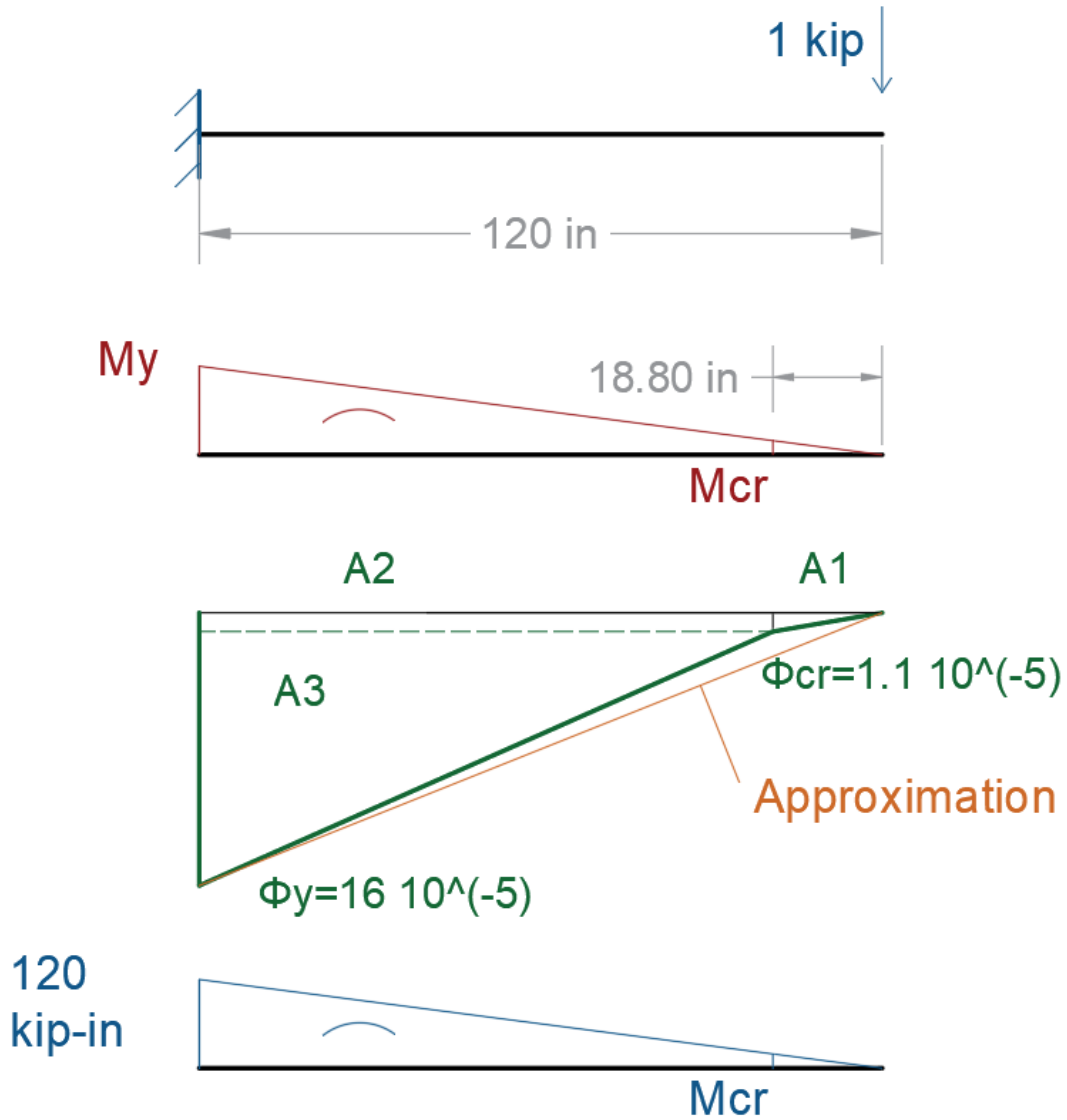
### 4.2 Comment on Flexure

The difference in results yield:

| Parameter               | Description                                                          | Value / Calculation | Reference                             |
|-------------------------|----------------------------------------------------------------------|---------------------|---------------------------------------|
| $A$                     | Unique area, triangular region depicted in orange in <b>Fig. 4</b> . | 0.0096              | From curvature diagram integration.   |
| $m_x$                   | Moment $m_x$ value at area centroid.                                 | 80 kip–in           | Virtual moment distribution.          |
| $\delta_f$              | Total flexural displacement (fully cracked assumption).              | <b>0.768</b> in     | $A \cdot m_x$                         |
| $\delta_v$              | Shear displacement.                                                  | 0.02019 in          | Based on effective shear modulus.     |
| $\delta_s$              | Slip displacement.                                                   | 0.00000 in          | No slip considered due to assignment. |
| $\delta_{\text{total}}$ | Total displacement.                                                  | <b>0.788</b> in     | $\delta_f + \delta_v + \delta_s$      |

**Table 5:** Flexural displacement under the fully cracked section assumption.

As observed, the value reported in **Table 5** is higher than that in **Table 4**. This supports the validity of the results, since **Fig. 4** clearly illustrates that the area under the moment–curvature curve along the beam span is larger.



**Figure 4:** Bending moment and curvature diagram along beam span. Cracked section simplification.

## 5 Problem d: Effect of slip displacement.

### Assignment 4

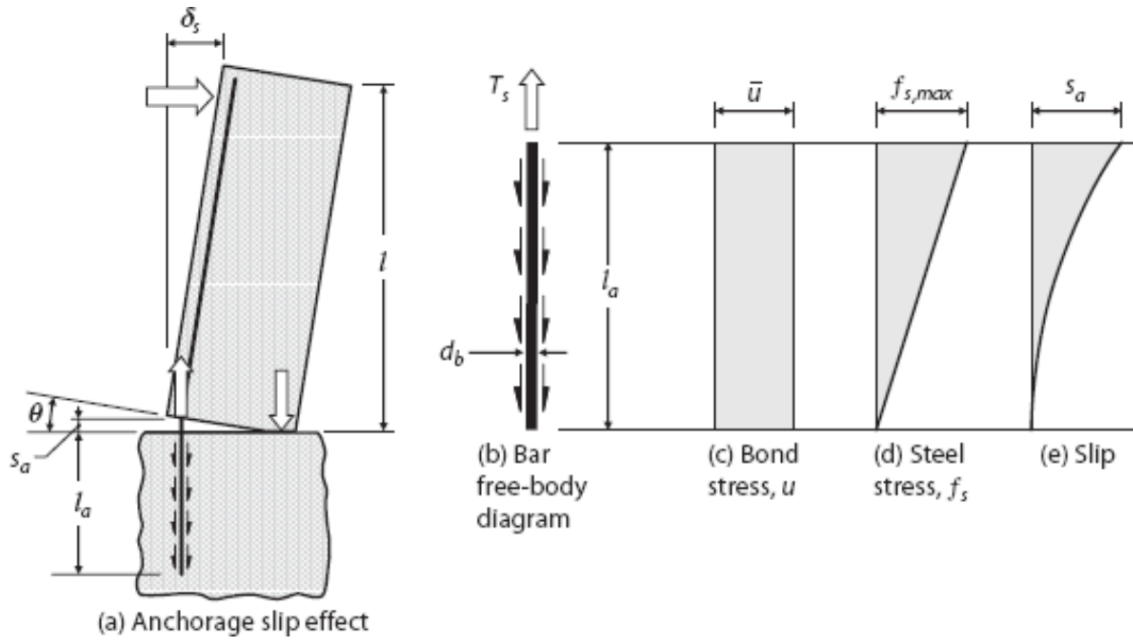
Repeat the previous problem including the effect of slip from the anchorage. Assume average uniform bond stress within the anchorage equal to  $12\sqrt{f'_c}$  psi. Do not forget that  $f_y = 69$  ksi.

### 5.1 Comment on Slip Displacement

In addition to flexural and shear deformations, longitudinal reinforcement slip at the anchorage can introduce further flexibility in the member. This phenomenon arises from the gradual transfer of tensile force from the steel to the surrounding concrete through bond stress along the anchorage length. As the bar elongates, the accumulated slip produces a rigid-body rotation of the member, leading to an additional tip displacement. This mechanism is illustrated in **Fig. 5**, where the slip-induced rotation  $\theta$  generates a corresponding displacement  $\delta_s = \theta l$  at the free end of the cantilever. The following section develops the calculation of this displacement component.

| Parameter           | Description / Formula                                                                | Value / Calculation | Reference                                    |
|---------------------|--------------------------------------------------------------------------------------|---------------------|----------------------------------------------|
| $d_b$               | Bar diameter.                                                                        | 1.0 in              | Figure 1                                     |
| $\bar{u}$           | Bond stress, $12\sqrt{f'_c}$ .                                                       | 758.95 psi          | Recommendation                               |
| $l_a$               | Anchorage length: $l_a = \frac{f_{s,\max}d_b}{4\bar{u}}$                             | 22.73 in            | Eq. (6.43)[6]                                |
| Anchorage condition | Verification of bar fit within anchorage zone.                                       | Verifies.           | The problem asks to consider infinite space. |
| $s_a$               | Stretch in traction rebar, Eq. (6.45): $s_a = \frac{f_{s,\max}^2 d_b}{8E_s \bar{u}}$ | 0.02704 in          | Under linear stress diagram[6].              |
| $k$                 | Factor for finding neutral axis depth.                                               | 0.305               | From section analysis                        |
| $d$                 | Distance from most compressive fiber to tension rebar.                               | 21.5 in             | Given                                        |
| $c$                 | Neutral axis depth.                                                                  | 6.56 in             | From section analysis                        |
| $\theta$            | Slip angle, Eq. (6.46): $\theta = \frac{s_a}{d - c}$                                 | 0.00181 rad         | Computed                                     |
| $\delta_s$          | Slip displacement, $\delta_s = \theta L$                                             | 0.21713 in          | From slip rotation                           |

**Table 6:** Computation of anchorage slip and resulting displacement using Eqs. (6.43)–(6.46).



**Figure 5:** Bond-slip model for straight bar anchorage[6].

Appending this results to the one obtained in Section 5 yields:

$$\delta_T = \delta_f^{PIII} + \delta_v + \delta_s = 0.768 \text{ in} + 0.002019 \text{ in} + 0.217 \text{ in} = 1 \text{ in}$$

The incorporation of the slip signifies:

$$\frac{\delta_s}{\delta_T} = \frac{0.217}{1} = 21.6\% \quad (5)$$

This percentage will be further discussed in Section 7.

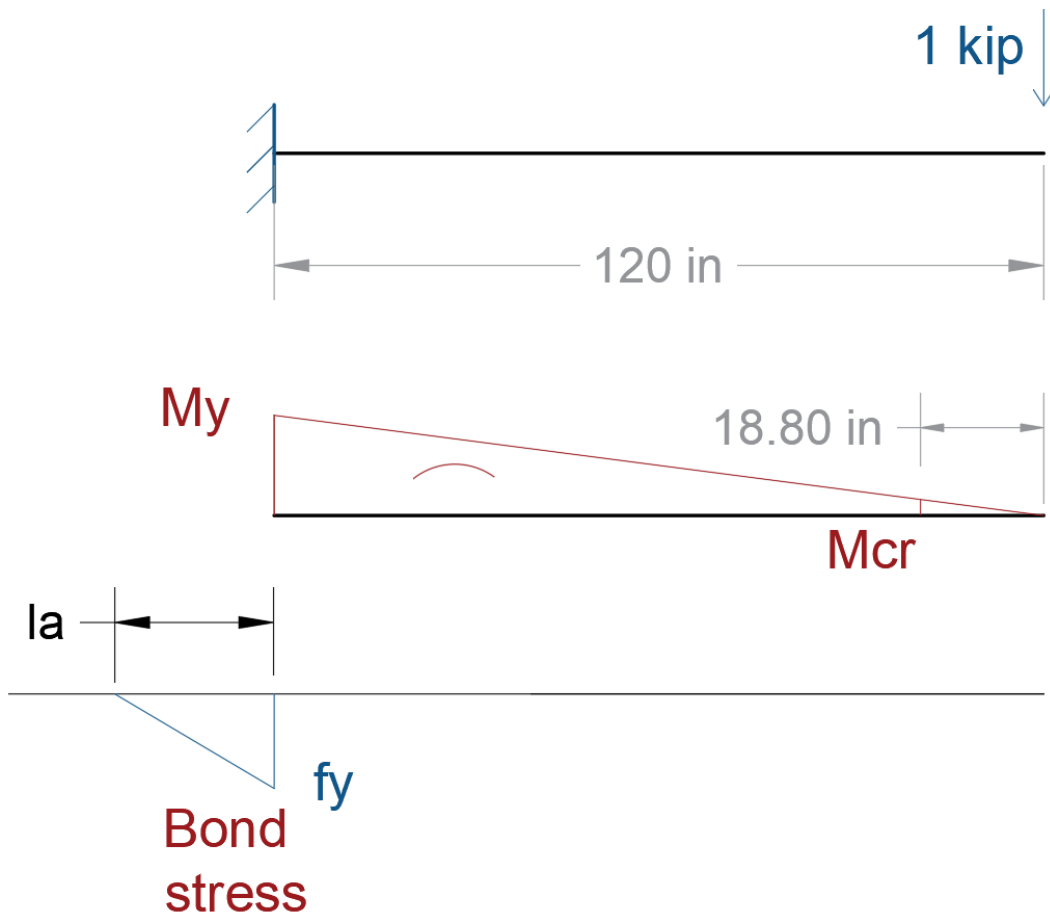


Figure 6: Anchorage length. Assumed linear stress variation.

## 6 Problem e: Plastic Hinge Model.

### Assignment 5

Use the plastic hinge model to calculate the displacement at which spalling occurs. Assume spalling occurs at compressive strain of 0.004. Use the plastic hinge model with  $l_p = h/2$ . Note that for the plastic hinge model, the displacement is the sum of a yield displacement and a displacement due to plastic hinge rotation. For this purpose, the yield curvature is extrapolated to the ultimate moment and slip displacements are not explicitly included (they are built implicitly into the plastic hinge model). You may also ignore shear deformations.

### 6.1 Preface

For the present analysis, all deformations were evaluated within the linear elastic range, prior to the onset of yielding. The plastic-hinge model, illustrated in ??, idealizes the member response through an elasto-plastic moment–curvature relation, in which the linear portion represents the pre-yield behavior. Since the section under study has not yet reached either the yielding of the tensile reinforcement or the onset of concrete crushing, the plastic region has not been activated. Therefore, no plastic curvature  $\phi_u - \phi_y$  or associated hinge rotation is considered at this stage.

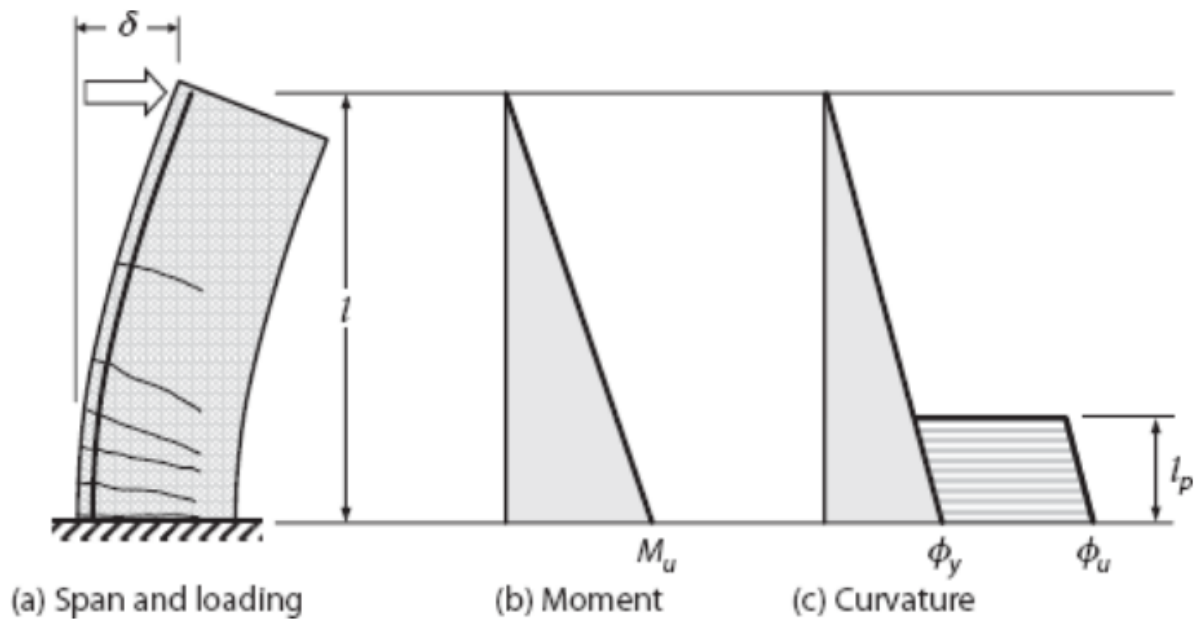


Figure 7: Plastic Hinge Model[6].

6.2 Results

| Parameter   | Description / Formula                                                | Value / Calculation                    | Reference                             |
|-------------|----------------------------------------------------------------------|----------------------------------------|---------------------------------------|
| $\phi'_y$   | Curvature if linear behavior continued, $\frac{M_{sp}}{M_y}\phi_y$ . | $0.00016397 \text{ in}^{-1}$           | From moment-curvature relation        |
| $\delta'_y$ | Corresponding displacement, $\phi'_y \frac{L^2}{3}$ .                | $0.78705 \text{ in}$                   | From elastic deflection theory        |
| $l_p$       | Plastic hinge length, assumed $h/2$ .                                | $12 \text{ in}$                        | Assumed following Eq. (6.59)[6]       |
| $\theta_p$  | Plastic rotation, $\theta_p = (\phi_u - \phi'_y)l_p$ .               | $0.01243 \text{ rad}$                  | Eq. (6.55)[6]                         |
| $\delta_p$  | Plastic displacement.                                                | $1.41729 \text{ in}$                   | Computed from $\theta_p$ contribution |
| $\delta_t$  | Total displacement, $\delta_t = \delta_p + \delta'_y$ .              | <b><math>2.20434 \text{ in}</math></b> | Incorporates all effects              |

**Table 7:** Computation of plastic hinge rotation and total displacement using the plastic hinge model (Eqs. 6.55–6.59).

## 7 Problem f: Results Comparison

### Assignment 6

Parts two and three were identical except for consideration of uncracked sections along the length. Did the final answer vary much with the method used? Parts three and four were identical except for inclusion of bar slip. Did the final answer vary much with the method used?

### 7.1 Consideration of Uncracked Bar Section

The results of Section 4.2 are expressed again for convenience:

$$\frac{\delta_T^c - \delta_T^b}{\delta_T^b} = \frac{0.788\text{in} - 0.749\text{in}}{0.749\text{in}} = 5.2\%$$

Notice that the superscript indicates the problem from where the total displacement was obtained. As was mentioned earlier, the value reported in **Table 5** is higher than that in **Table 4**. This supports the validity of the results, since **Fig. 4** clearly illustrates that the area under the moment–curvature curve along the beam span is larger. It is concluded that, while significant, this value represents a smaller effect in the final displacement from a conservative viewpoint.

### 7.2 Bar slip effect

Eq. (5), is brought for convenience:

$$\frac{\delta_s}{\delta_T^d} = \frac{0.217}{0.987} = 22\%$$

The inclusion of bar slip generates a 22% in the total result of the displacement, which is considerable. The accumulated slip produces a rigid-body rotation of the member, leading to an additional (and considerable) tip displacement[6].

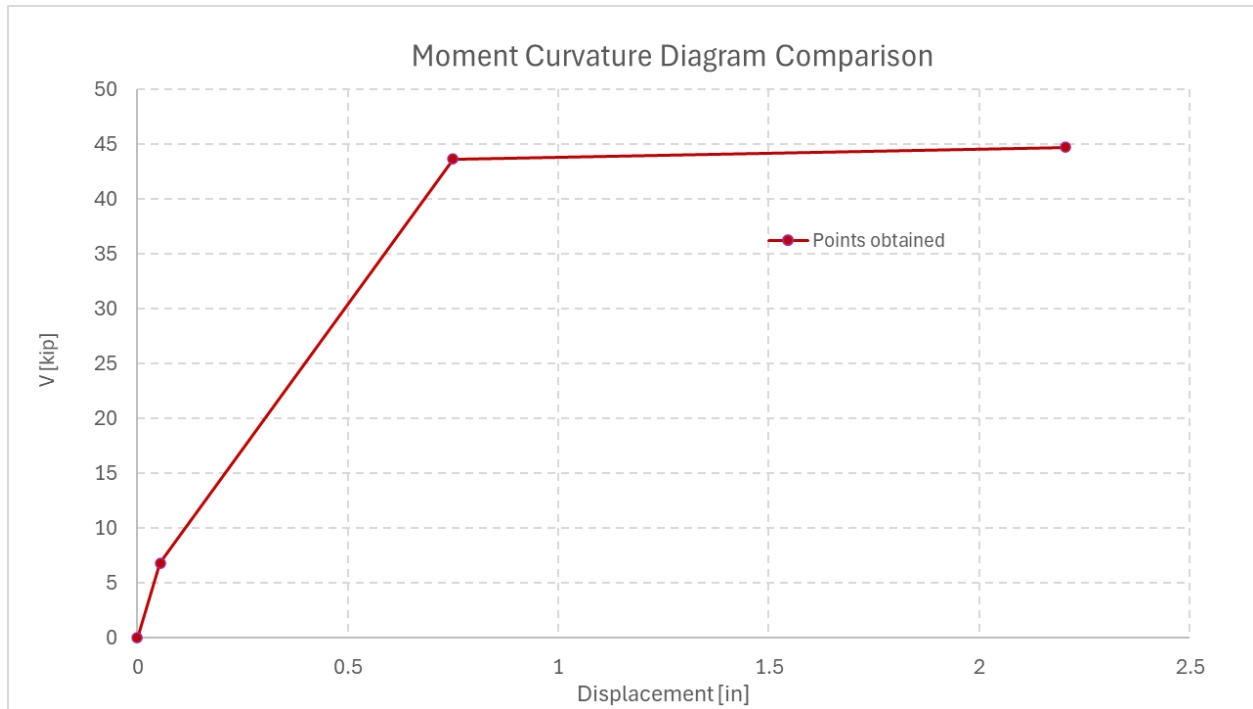
### Assignment 7

Plot your best estimate of the load–displacement relationship, that is, a plot starting from the origin, through cracking, from there through yield including slip, and from there to ultimate.

**Table 8** summarizes the complete load–displacement relationship of the cantilever beam and it is plotted in **Fig. 8**. It consolidates the results obtained from each stage of the analysis, including the elastic response prior to cracking, the linear behavior up to yielding with cracked and uncracked sections with slip effects, and the ultimate response governed by the plastic hinge formation. The table clearly reflects the progressive increase in displacement associated with each structural stage.

| Displacement [in] | Shear Force [kip] | Description                                | Reference                                         |
|-------------------|-------------------|--------------------------------------------|---------------------------------------------------|
| 0.00000           | 0.00              | Initial unloaded state.                    | Reference position.                               |
| 0.05438           | 6.83              | At onset of cracking.                      | From uncracked section analysis in Section 2.     |
| 0.74922           | 43.67             | At yielding of longitudinal reinforcement. | From linear analysis including slip in Section 5. |
| 2.20434           | 44.75             | At spalling / ultimate stage.              | From plastic hinge model in in Section 6.         |

**Table 8:** Summary of the load–displacement relationship for the cantilever beam.



**Figure 8:** Load-displacement relationship.

## 8 References

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